

Structure and dynamics of a Gay–Berne liquid crystal confined in cylindrical nanopores

Qing Ji, Ronan Lefort, Rémi Busselez, and Denis Morineau

Citation: *J. Chem. Phys.* **130**, 234501 (2009); doi: 10.1063/1.3148889

View online: <http://dx.doi.org/10.1063/1.3148889>

View Table of Contents: <http://jcp.aip.org/resource/1/JCPSA6/v130/i23>

Published by the AIP Publishing LLC.

Additional information on J. Chem. Phys.

Journal Homepage: <http://jcp.aip.org/>

Journal Information: http://jcp.aip.org/about/about_the_journal

Top downloads: http://jcp.aip.org/features/most_downloaded

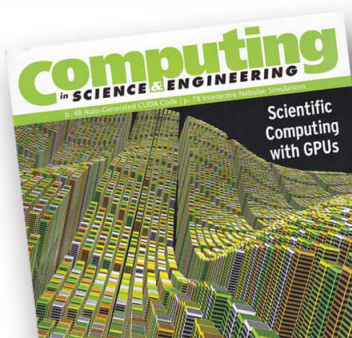
Information for Authors: <http://jcp.aip.org/authors>

ADVERTISEMENT

**SHARPEN YOUR
COMPUTATIONAL
SKILLS.**



Subscribe for
\$49 | year



computing
in **SCIENCE & ENGINEERING**

Scientific
Computing
with GPUs

Structure and dynamics of a Gay–Berne liquid crystal confined in cylindrical nanopores

Qing Ji, Ronan Lefort, Rémi Busselez, and Denis Morineau^{a)}

Institute of Physics of Rennes, CNRS-University of Rennes 1, UMR 6251, F-35042 Rennes, France

(Received 28 October 2008; accepted 14 May 2009; published online 15 June 2009)

Gay–Berne liquid crystals confined in two cylindrical nanopores with different pore sizes were studied by molecular dynamics simulation. Their structure and dynamics properties were obtained and compared with those of the bulk. Our data show that confinement changes the bulk isotropic-to-nematic transition to a continuous ordering from a paranematic to a nematic phase. Moreover, confinement strongly hinders the smectic translational order. The molecular dynamics is characterized by the translational diffusion coefficients and the first-rank reorientational correlation times. Very different characteristic times and temperature variations in the dynamics are observed in confinement. Spatially resolved quantities illustrate that confinement induces predominant structural and dynamical heterogeneities. © 2009 American Institute of Physics. [DOI: 10.1063/1.3148889]

I. INTRODUCTION

Liquid crystals (LCs) in confined geometry are extensively utilized for various technological applications, including displays, optical switches, and many other electro-optical devices. Moreover, the physics of confined LCs reveal several original features, many of which are still open questions fundamentally, such as the theoretical treatment of phase behaviors under external fields induced by confinement or the interfacial properties of soft-matter and biomimetic materials. Therefore, many experimental and theoretical studies focused on this scientific case during the last decade.^{1–10} Simple porous geometries, in terms of unidirectional arrays of straight nanochannels of silicon and silica, have been recently highlighted as unprecedented systems in order to gain a deeper insight into the isotropic to nematic (*I*–*N*) and nematic to smectic *A* (*N*–Sm*A*) phase transitions of spatially confined LCs.^{11–16}

Under confinement, the surface interaction acts as an external field that couples to the LCs order parameters. This effect may induce partial order at the pore wall, which persists well above the clearing temperature.^{2,3} Moreover the phase behaviors are considerably affected by spatial confinement and interface effects as well. For instance, the *I*–*N* transition is weakened and its character changes from a strong-first order transition in bulk to a weak first-order or continuous paranematic-to-nematic (*P*–*N*) transition depending on the strength of surface interaction.^{4,14} The smectic translational order is also inhibited and the *N*–Sm*A* is suppressed or considerably broadened.^{5,10,12}

Another important issue of current interest concerns the average dynamics of LC particles in terms of translational diffusion and molecule axial rotation. Both of them are also strongly modified in confinement. Dielectric spectroscopy and quasielastic neutron scattering (QENS) experiments revealed a significant slowdown of the relaxation

modes.^{6,8,9,15,16} The signature of bulk phase transitions on dynamics is often smeared out in confinement. The observed features exhibit some similarities with glass-forming systems, such as a non-Debye character of the relaxation decay and a non-Arrhenius temperature dependence of the relaxation time.^{6,8,9,16} Recent QENS experiments on LCs in silicon nanochannels suggested that the interfacial slowing down effect due to the hindered rotation and diffusion of the surface molecules propagates into the pore volume, leading to spatially strongly heterogeneous dynamics.^{15,16}

At variance to most experiments, molecular simulation can directly scrutinize the properties of the confined LCs with a spatial resolution at the molecular scale.¹⁷ It is a unique method allowing one to gain a better insight into the ordering and dynamics of LCs in nanopores. Single site anisotropic molecular potentials derived from the initial Gay–Berne model (GB) have shown to contain the essential features of the intermolecular interaction between LC mesogens.¹⁸ Each GB model is defined by a set of four parameters κ , κ' , μ , and ν , which allows one to tune the shape and especially the anisotropy of the interaction energy. The parametrization considered by Gay and Berne¹⁸ in their original work (3, 5, 2, 1) has been extensively studied.^{19–23} Luckhurst and Simmonds²⁴ mapped the GB interaction form to the p-terphenyl molecule and derived a more anisotropic model, characterized by larger parameters (4.4, 39.6, 0.8, 0.74). Following this work, Bates and Luckhurst²⁵ proposed a modified parametrization (4.4, 20, 1, 1), where the exponents μ and ν were both set equal to unity for computational convenience. At variance to the original GB parametrization, the higher anisotropic character of the latter model stabilizes the smectic *A* (Sm*A*) phase, in addition to the isotropic (*I*), nematic (*N*), and crystal (Cr) phases. This parametrization corresponds to one of the most studied GB model so far.^{25–31} It has been adopted in the present study also in order to better match our recent experimental studies of smectogens.^{12–16}

Molecular simulations of LCs in confined geometry have

^{a)}Electronic mail: denis.morineau@univ-rennes1.fr.

been mostly used in the slab geometry. They investigated the microscopic structure of LCs in the vicinity of an interface and the parameters that control both the first layer tilt and LC pretilt angle with respect to standard continuum theory.^{32–36} Variable temperature or pressure molecular simulations in the slab geometry have been performed with the GB model (or a simplified version of it) with the original parameterization (3, 5, 2, 1).^{37–41} In qualitative agreement with experiments, these simulations found that the confinement induces a significant smearing of the transitions. Paranematic ordering is either favored or disrupted depending on the nature of the surface interaction and surface roughness. Meanwhile, isothermal simulations reported that the phase transitions are either shifted to higher or lower density.⁴¹ Although there are some advantages of a higher level of simplicity, which allows for better theoretical handling, it is noteworthy that the slab geometry does not compare with most of the experiments using porous solids as confining media. The porous solids (cylinder, spheres, or interconnected cavities) are indeed likely to induce stronger topological and elastic constraints to the mesomorphic phases and are less subject to peculiar commensuration effects than slab pores.^{5,42} The case of cylindrical or spherical pores has been generally addressed by simulation studies of simplest models, such as lattice models or hard spherocylindrical particles.^{43,44} Reports from simulation studies of GB systems in curved porous geometries are very rare in the literature, such as one study on GB(3,5,2,1) droplets by Emerson and Zannoni.⁴⁵ Given the number of recent experimental results on LCs confined in nanochannels, simulations of smectics in cylindrical pores are particularly timely.^{8,12–16} More striking, until the recent study by Mima and Yasuoka,⁴⁶ the dynamical properties of LC molecules in confinement have only been investigated on a very limited basis by off-lattice molecular simulation. Again, invaluable information may be expected from molecular dynamics (MD) simulations, which could ultimately unravel the microscopic origin of the glasslike slowdown, the non-Debye type, and the non-Arrhenius temperature dependence of the dynamical properties of nanoconfined LCs, as reported by various experiments.^{6,8,9,15,16}

The aim of this manuscript is to report the results of a GB system confined in cylindrical nanopores, studied for the first time, by MD simulations. In this contribution, a description of both the structural and the dynamical properties will be presented as a function of the temperature and pore size.

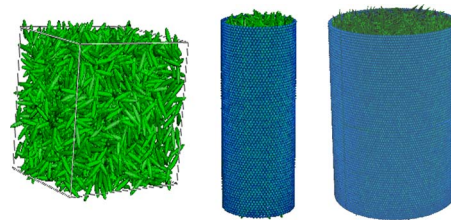


FIG. 1. (Color online) Snapshots of instantaneous configurations of the liquid-crystal in the isotropic phase. From left to right: bulk, smaller cylinder ($D20$), and bigger cylinder ($D40$).

II. THE MODELS

We simulated three LCs systems: The bulk and two systems confined in different cylindrical pores. The diameters of the two cylindrical pores are 20 and 40 times larger than the short diameter of a LC particle. The two confined systems will be subsequently denoted as $D20$ and $D40$, respectively. If one maps the short diameter of a LC particle in our simulation to that of a real LC molecule, typically 4 Å for 4-n-octyl-4-cyanobiphenyl, the corresponding pore sizes (diameters) are about 8 and 16 nm. A snapshot of these systems is shown in Fig. 1. The confinement effects are analyzed by a systematic comparison between the properties of the confined phases and those of the bulk.

The interactions between LC molecules are expressed by the pair potential proposed by Gay and Berne for axially symmetric particles, which can be written for two arbitrary molecules i and j , with unit vectors along their symmetry axis $\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j$ as¹⁸

$$U_{ij}^{\text{GB}}(\mathbf{r}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) = 4\varepsilon(\hat{\mathbf{r}}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) \left[\left(\frac{\sigma_0}{d(\mathbf{r}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j)} \right)^{12} - \left(\frac{\sigma_0}{d(\mathbf{r}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j)} \right)^6 \right], \quad (1)$$

where $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$ is the unit vector along the vector $\mathbf{r}_{ij}$ joining the molecules i and j , and σ_0 defines the smallest molecular diameter. The form of the GB potential arises from a generalization of the expression of the Lennard-Jones (LJ) potential, which is recovered exactly in the case of spherical particles. The anisotropic effect in the interaction is introduced in the energetic term $\varepsilon(\hat{\mathbf{r}}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j)$ and the distance $d(\mathbf{r}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j)$ as follows:

$$\varepsilon(\hat{\mathbf{r}}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) = \varepsilon_0 \left[1 - \left(\frac{\kappa^2 - 1}{\kappa^2 + 1} \right)^2 (\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2 \right]^{-\nu/2} \left\{ 1 - \left(\frac{\kappa' - 1}{\kappa' + 1} \right) \left[\frac{(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}}_{ij})^2 + (\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})^2 - 2 \left(\frac{\kappa' - 1}{\kappa' + 1} \right) (\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}}_{ij})(\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)}{1 - \left(\frac{\kappa' - 1}{\kappa' + 1} \right)^2 (\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2} \right] \right\}^{\mu}, \quad (2)$$

$$d(\mathbf{r}_{ij}, \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) = r_{ij} + \sigma_0 - \sigma_0 \left[1 - \left(\frac{\kappa^2 - 1}{\kappa^2 + 1} \right) \left(\frac{(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}}_{ij})^2 + (\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})^2 - 2 \left(\frac{\kappa^2 - 1}{\kappa^2 + 1} \right) (\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}}_{ij})(\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)}{1 - \left(\frac{\kappa^2 - 1}{\kappa^2 + 1} \right)^2 (\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2} \right) \right]^{-1/2}. \quad (3)$$

These equations are parametrized by four variables, κ , κ' , ν , and μ . The Greek term κ stands for the length-to-breadth ratio of the molecule, while κ' stands for the ratio of the potential well depths between the side-by-side and the end-to-end configurations. The exponents ν and μ are adjustable parameters that have been introduced to tune the shape of the anisotropic interaction, by affecting the nematic and smectic ordering ability of the particles in a subtle way.²⁵ We set the four parameters to 4.4, 20, 1, and 1, respectively. These values correspond to one of the most studied model throughout the literature.^{25–31} At variance to the original parametrization (3, 5, 2, 1), which has no stable SmA phase, the simulated LC bulk with those parameters listed above exhibits I, N, SmA, and Cr phases and is more similar to recent experiments.^{8,12–16,21,26}

The pore wall was created by decorating the surface of the cylinders with LJ atoms in close contact and arranged on a triangular array. The LJ parameters corresponding to energy and particle size are set equal to the GB side-to-side interaction, i.e., $\varepsilon_{\text{LJ}} = \varepsilon_0$ and $\sigma_{\text{LJ}} = \sigma_0$. Since the wall particles are spherical, their interaction with the GB particles depends only on the distance between the two interaction sites and the orientation of the GB particle axis $\hat{\mathbf{u}}_j$. GB-LJ interactions were computed from Eq. (4) derived by analogy to GB-GB interactions,¹⁸ by setting $\hat{\mathbf{u}}_i$ equals to zero in Eqs. (1)–(3) as follows:

$$U_{ij}^{\text{GB-LJ}}(\mathbf{r}_{ij}, \hat{\mathbf{u}}_j) = 4\varepsilon^{\text{GB-LJ}}(\hat{\mathbf{r}}_{ij}, \hat{\mathbf{u}}_j) \left[\left(\frac{\sigma_0}{d^{\text{GB-LJ}}(\mathbf{r}_{ij}, \hat{\mathbf{u}}_j)} \right)^{12} - \left(\frac{\sigma_0}{d^{\text{GB-LJ}}(\mathbf{r}_{ij}, \hat{\mathbf{u}}_j)} \right)^6 \right], \quad (4)$$

with

$$\varepsilon^{\text{GB-LJ}}(\hat{\mathbf{r}}_{ij}, \hat{\mathbf{u}}_j) = \varepsilon_0 \times \left\{ 1 - \left(\frac{\kappa' - 1}{\kappa' + 1} \right) (\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})^2 \right\}^\mu, \quad (5)$$

and

$$d^{\text{GB-LJ}}(\mathbf{r}_{ij}, \hat{\mathbf{u}}_j) = r_{ij} + \sigma_0 - \sigma_0 \left[1 - \left(\frac{\kappa^2 - 1}{\kappa^2 + 1} \right) (\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}}_{ij})^2 \right]^{-1/2}. \quad (6)$$

III. METHOD AND PARAMETERS

We performed MD simulation using the LAMMPS software package, developed at Sandia National Laboratories, in the canonical *NVT* ensemble.⁴⁷ All quantities are expressed in conventional reduced units, with σ_0 and ε_0 being the units of length and energy, respectively. Thus, the temperature is given in units of ε_0/k_B , the pressure in units of ε_0/σ_0^3 , the

momentum of inertia in units of $m\sigma_0^2$, and the number density ρ in units of $1/\sigma_0^3$. The value of the momentum of inertia was set to $I_\perp = 1/5(1 + \kappa^2)$, which corresponds to the value for an homogeneous ellipsoidal particle of axis 1, 1, and κ . The Nose–Hoover temperature thermostat is applied and the integration time step is set to 0.002. Like previous studies, the intermolecular interactions were truncated at $r_c = 5.5\sigma_0$ and no long-range corrections were applied.²⁶ Periodic boundary conditions were applied in the three dimensions for the bulk systems and along the pore channel z -axis for the confined systems.

It is relevant to stress that the constant *NVT* ensemble may not appear as the most suitable ensemble to investigate the phase transition. Indeed, it avoids the observation of true first order transitions along constant density paths. Grand canonical or isobaric-isothermal ensembles would appear more suitable for this goal. However, box shape fluctuations are hardly possible in the situation of rigid confinement, which is one of interest to our group. Therefore, as widely seen in many literature reports on LC studies, *NVT* constant volume simulations have been chosen in the present study for simplicity.^{38,39,43,48,49} Therefore, we expect to gain information from the analysis of the structural and dynamical changes induced by confinement with respect to the bulk system along the same isochoric path.

A preliminary constant pressure study (results not shown) has been performed with the ultimate goal to check the validity of our method. The simulation of a system comprising 2400 particles was performed during heating and cooling at constant pressure $P=2.0$ in a temperature range from $T=0.8$ to $T=1.9$. The results obtained from this study were in quantitative agreement with de Miguel *et al.*²⁶

Constant density simulation at $\rho=0.153$ was performed in the bulk system, which consists of 2400 particles in a rectangular a rectangular box of cell dimension $L_x=27.4$, $L_y=21.8$, and $L_z=26.2$. The density $\rho=0.153$ is the equilibrium value of the bulk simulation with $P=2.0$ and $T=1.9$. Confined systems at the same density comprise $N_{20}=2885$ and $N_{40}=11\,544$ particles in cylindrical pores with diameters 20 and 40, respectively, while the pore length for both cylinders was $L_z=60$. Isotropic starting configurations were obtained after the full melting and thermalization of initial crystalline arrangements, obtained after 300 000 steps of *NVT* simulation at temperature $T=1.9$. Similar thermal treatments were subsequently applied to the three samples. They were cooled down with temperature change interval of 0.05. At each temperature, we sequentially ran 60 000 cycles for equilibration and 60 000 cycles for data collection and averaging.

The structure of the different phases was characterized

by one orientational order parameter (S) and one translational order parameter (τ). The orientational order parameter S is defined from Eq. (7), where $\mathbf{n}$ is the nematic director, $\hat{\mathbf{u}}_i$ is the unit vector along the principal axis of the molecule i , and $\langle \rangle_i$ stands for the average over the particles i .

$$S = \langle 3(\hat{\mathbf{u}}_i \cdot \mathbf{n})^2 - 1 \rangle_i / 2. \quad (7)$$

In an anisotropic system, the director $\mathbf{n}$ is the unit vector, which maximizes the value of Eq. (7). It can be defined as the eigenvector of the largest eigenvalue S of the ordering tensor $\mathbf{Q}$, whose components are $Q_{\alpha\beta} = \langle 3u_{i\alpha}u_{i\beta} - \delta_{\alpha\beta} \rangle_i / 2$. S is a measure of the degree of orientational order of the phase. It may vary from 0 for an orientationally disordered system to 1 for a perfectly uniaxially oriented system.

In the confined system $D40$, we also introduced a planar-tangential order parameter $S_T = \langle 3(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}}_i^\perp)^2 - 1 \rangle_i / 2$, where $\hat{\mathbf{r}}_i^\perp$ is the unit vector contained within the xy -plane and perpendicular to the radial vector that connects the particle center to the pore axis ($\hat{\mathbf{r}}_i^\perp$ is tangential to the pore surface).

The translational order parameter τ is related to the static structure factor along the direction of the director for the wave vector $q = 2\pi/d$ and is defined by

$$\tau = \left| \left\langle \exp\left(i \frac{2\pi}{d} z_i\right) \right\rangle_i \right|. \quad (8)$$

In Eq. (8), d is the interlayer spacing and z_i is the ordinate of the center of the particle i along an axis parallel to the director. The interlayer spacing d is not known *a priori* and is rigorously defined as the distance which maximizes Eq. (8). The final interlayer spacing was obtained by testing a set of values of d until a maximum value for τ was obtained. The order parameter τ is a measure of the onset of a translational symmetry, which results from a periodic modulation of the density correlation functions. It may vary from 0 to 1 and is especially suitable to detect the layered structures that occur in the smectic phases.

IV. STRUCTURE OF THE CONFINED GB

The orientational order parameters S of the three samples measured on cooling from $T = 1.9$ to $T = 0.8$ are shown in Fig. 2(a). For the bulk system, the observation of a nonzero value of the order parameter S in the isotropic phase at $T = 1.9$ is ascribed to a well-known finite size effect.²⁶ A rapid increase in S occurs at about $T = 1.4$, which has been attributed to the isotropic-to-nematic phase transition ($I-N$). It should be noted that the increase in S is sharp, but not discontinuous. This is a direct effect of the constant volume procedure used in the simulations, which avoids the discontinuous density change expected for a true first-order $I-N$ phase transition.

The orientational order parameter of the LC confined in cylinder $D20$ displays two striking features. First, its value is already large at high temperatures where the bulk system is isotropic ($S = 0.35$ at $T = 1.9$). This feature shows that cylindrical confinement promotes a nematic ordering well above the bulk $I-N$ phase transition. This effect has been analyzed experimentally and theoretically and is usually denoted as paranematic ordering.^{8,14,50} It is ascribed to the effect of the unidirectional field induced by the confinement and the in-

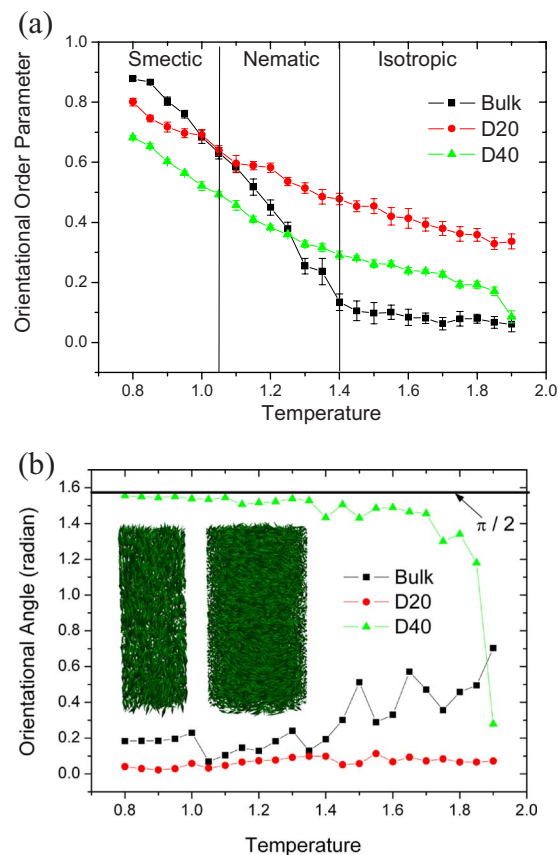


FIG. 2. (Color online) (a) Temperature variation in the orientational order parameter of the GB systems on cooling at constant density $\rho = 0.153$ in bulk (squares) and in two cylinders $D20$ (circles) and $D40$ (triangles). The temperature regions of the bulk phases are recalled. (b) Temperature variation in the angle formed by the averaged nematic director with respect to the z -axis. Same symbols are signified as (a). In the inset are snapshots of $D20$ and $D40$ configurations at $T = 1.2$.

teraction with the surface, which couples to the nematic order parameter. Second, the nematic ordering increases gradually on cooling and reaches a value of $S = 0.8$ at $T = 0.8$, which is typically observed in the bulk nematic phase. However, there is no indication of any jump or kink in its temperature variation that would reflect the occurrence of a regular $I-N$ transition. The gradual increase in the nematic order on cooling indicates that the surface interaction transforms the bulk discontinuous $I-N$ transition to a gradual paranematic-nematic ($P-N$) ordering. The effect of surface interaction on the nature of the $P-N$ transition in confined geometry has been discussed by Kutnjak *et al.*⁵⁰ in the frame of a model derived from the Landau-de Gennes theory. This theoretical approach predicts the existence of a critical threshold value of the strength of the orientational ordering field, which separates discontinuous from continuous $P-N$ transition. This result is also in agreement with very recent experimental observations of a continuous $P-N$ transition for cyanobiphenyls confined in 10 nm silica channels, which exhibit a similar temperature variation in the orientational order parameter.¹⁴ The orientational order parameter of the LC confined in cylinder $D40$ is always slightly lower than that of $D20$. They share a similar temperature variation despite different structures of the nematic director.

Indeed, Fig. 2(b) reveals that the orientation of the direc-

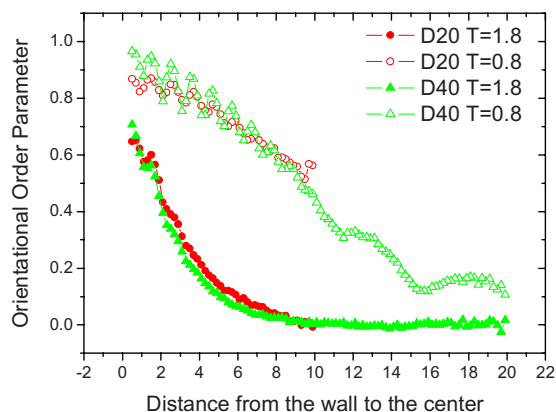


FIG. 3. (Color online) The radial profile of the spatially resolved orientational order parameter of the two confined GB systems at two different temperatures. The order parameter directions are axial and planar-tangential for *D20* and *D40*, respectively.

tor for cylinder *D20* lies along the pore axis for temperature below $T=1.8$, which corresponds to the axial-homogeneous nematic structure. The structure adopted by LCs in confinement primarily depends on the anchoring condition, which is ascribed to the balance between molecule-wall and molecule-molecule interaction. The model used in this study promotes planar anchoring. In addition to the anchoring strength, one must consider the energetic contribution that would require other structures to form in cylindrical geometry in terms of elastic distortion or topological defects. The axial-homogeneous configuration observed for *D20* is the only one, which assumes a planar anchoring and allows the growth of a defect-free nematic domain with negligible elastic deformation. On the contrary, the nematic director in *D40* lies within the *xy*-plane and corresponds to a preferred orientation of the molecules long axis that is tangential to the pore surface as shown in the Fig. 2(b). Therefore, the planar-tangential local order parameter S_T as defined in Sec. III is more appropriate for cylinder *D40*. The observation of the planar-tangential configuration in *D40* is surprising and probably related to the combination of the strong planar surface interaction and the small value of the pore curvature. In the case of a very large pore, as studied here, both planar-axial and planar-tangential configurations have comparable energy and are degenerated structures for flat surfaces (vanishing curvature). According to this, the existence of different planar metastable configurations of comparable energy is possible if sufficiently large pore diameters are considered.

Interestingly, the radial distribution of the orientational order parameters as a function of the distance to the wall are essentially the same for *D20* and *D40* systems, as shown in Fig. 3. This signifies that, with the set of parameters used in this study, the size of *D40* exceeds the threshold beyond which curvature effects do not affect significantly the nematic order induced by the surface. The surface order parameter is high for the two cylinders whatever the azimuthal orientation or temperature. However, the nematic order parameter in the center of *D40* remains small at $T=0.8$ and is null at $T=1.8$. This situation is different for *D20*, which shows a large nematic order at the pore center. The latter case is very unfavorable to a nonhomogeneous arrangement

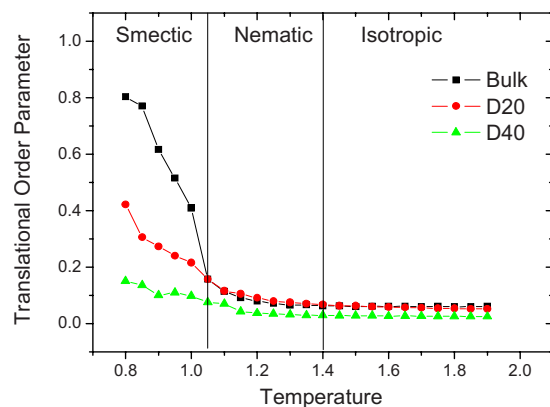


FIG. 4. (Color online) Temperature variation in the translational order parameter of the GB systems on cooling at constant density $\rho=0.153$ in bulk (squares) and in two cylinders *D20* (circles) and *D40* (triangles).

such as planar-tangential, which is unstable for small radii. This central nematic domain probably drives the crossover seen between the two different arrangements.

The translational order parameters recorded along the same cooling scans are displayed in Fig. 4. For the cylinder *D20*, this order parameter is bulklike in the whole high temperature region ($T > 1.05$). Below $T=1.05$, the smectic order increases slowly in cylinder *D20*, indicating a smearing of the nematic-to-smectic (*N-SmA*) transition compared to that of the bulk. Strong commensuration effects leading to enhanced smectic ordering in slit pores have been predicted by DFT calculations,⁴² such effects are less important in cylindrical pores or in other types of curved porous geometries. The situation encountered in our simulation probably shares some similarities with recent experiments on octylcyanobiphenyl confined in porous silicon straight channels.¹² In the experiments, the *N-SmA* transition is inhibited and replaced by the gradual growth of short range correlations with a homogeneous axial configuration. In addition, the planar-tangential configuration assumed by the GB particles in *D40* inhibits the formation of a homogeneous smectic domain.

V. DYNAMICS OF THE CONFINED GB

The MD of the GB model has been characterized by two standard quantities: The rotational correlation time τ_R and the mass center diffusion coefficient D_T .²² The reorientation of the GB molecules has been described by the first rank time correlation function of the unit vector defined as $C(t) = \langle \mathbf{u}(t+t_0) \mathbf{u}(t_0) \rangle$, where $\mathbf{u}(t)$ is the unit vector along the principle molecular axis at time t . This function characterizes the end-over-end motion of the molecules, which was probed by dielectric relaxation and IR spectroscopies for dipolar real LC mesogens.^{6,8,9} The rotational correlation time τ_R has been obtained for the reorientational correlation function assuming as a single exponential decay. The diffusion coefficient has been computed from the long time variation of the mean square displacement, according to

$$D_T = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \langle |\mathbf{r}(t+t_0) - \mathbf{r}(t_0)|^2 \rangle.$$

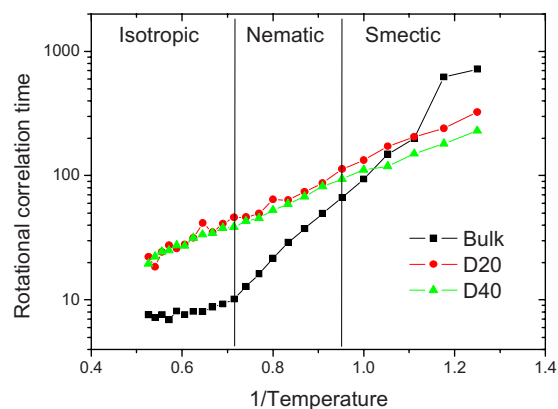


FIG. 5. (Color online) Variations in the rotational correlation time of the GB systems with the inverse temperature.

Figure 5 shows the rotational correlation time τ_R of the three GB systems on cooling in Arrhenius coordinates. For the bulk, τ_R displays a nearly Arrhenius behavior with a small activation energy in the isotropic phase. An abrupt change in slope in log scale occurs at the *I-N* transition. A second almost Arrhenius behavior with a much larger activation energy is then further observed in the nematic and smectic phases. It is at variance with the two confined GB systems, which display a unique Arrhenius behavior throughout the entire studied temperature range. Above the bulk *I-N* transition temperature ($1/T < 0.72$), the correlation time of the confined phase is up to four times larger than the bulk. The activation energies of τ_R in the confined phases are smaller than those of the bulk nematic and smectic phases. Consequently, the rotational dynamics of the confined phase is faster than the bulk for temperatures below about $T=1.0$.

The nematic ordering restricts the end-over-end molecular motion so that the variation in τ_R with temperature in the bulk GB system around $1/T \approx 0.72$ can be interpreted as a direct consequence of the abrupt increase of the nematic order parameter at the *I-N* transition. For the confined systems, the large slowdown of the rotational dynamics in the high temperature region can be related to the combination of surface hindered rotation and to surface induced paranematic ordering.^{6,8,9,16,46} The observation of the almost constant activation energy is related to the gradual character of *P-N* ordering in confinement and the absence of any other phase transition in the investigated temperature range.

The diffusion coefficients for the three GB systems on cooling are shown in Arrhenius coordinates in Fig. 6. In the isotropic and nematic phases, the diffusion coefficient of the bulk decreases gradually. It practically follows an Arrhenius temperature law with constant activation energy. The diffusion coefficient bends in the nematic phase on approaching the *N-SmA* transition. At lower temperature (in the smectic phase) it almost conforms to a second Arrhenius law with higher activation energy.

In confinement the translation diffusion is reduced by 1.3 in the isotropic phase for the two cylinders. Below $T=1.1$ ($1/T > 0.9$), the diffusion coefficient of *D20* displays a change of slope, which is reminiscent of the bulk system on approaching the *N-SmA* phase transition. It then further follows an almost Arrhenius temperature dependence, with an

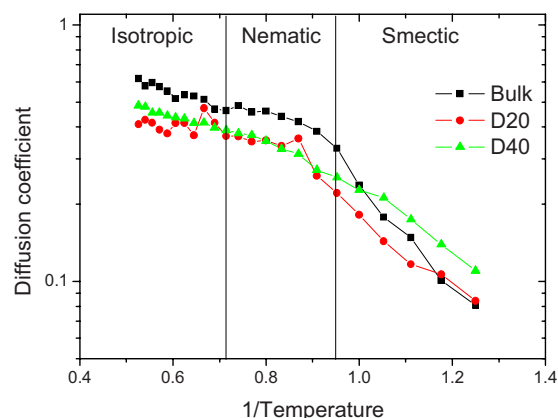


FIG. 6. (Color online) Variations of the diffusion coefficients of the GB systems with the inverse temperature.

activation energy smaller than the bulk. The case of *D40* is quite comparable to that of *D20* but with more gradual changes at low temperature. Indeed, the activation energy of *D40* seems less sensitive to the temperature, although it might depart from a single Arrhenius law too.^{6,8}

The translational dynamics are deeply reduced in the isotropic and nematic phases when the LCs are confined to cylinders. It is similar to previous observations on LCs and also globular liquids.^{15,16,46,51-54} This phenomenon can be attributed to a surface induced slowdown of the molecular mobility, which may depend on the surface interaction strength and surface roughness.⁵² This surface effect on the dynamics of the contact layer may propagate within the pore, changing the overall dynamics of the confined phase.^{16,51-53} On the other hand, the MD of the confined phase is affected by its structure also and faster dynamics is usually expected for a less ordered phase. In the previous section we have shown that the confined GB systems are less ordered than the bulk systems at low temperature. It is particularly obvious in the smectic phase as shown in Fig. 4. The coupling between structure and dynamics induces an additional acceleration effect on the MD. This effect is in competition with the surface induced slowdown initiated at high temperature. As a consequence, the increase in the activation energy is limited in confinement leading to an apparent acceleration of the dynamics at low temperature with respect to the bulk.

VI. IN-PORE STRUCTURAL AND DYNAMICAL HETEROGENEITIES

In confinement, the pore wall breaks the translation invariance of the system and induces spatial inhomogeneities within the porous volume.^{34,37,55} Direct inspection of such features is hardly possible experimentally although indirect signatures of surface-induced heterogeneities have been reported for many confined systems.^{6,8,16,54,51,56} Fortunately, this issue can be directly addressed by using molecular simulation method.

Figure 7 displays the radial distribution of the density ρ , the short-time angular rotation $\theta(\tau^*)$, and the short-time translation coefficient $D(\tau^*)$ in *D20* at $T=0.8$, with

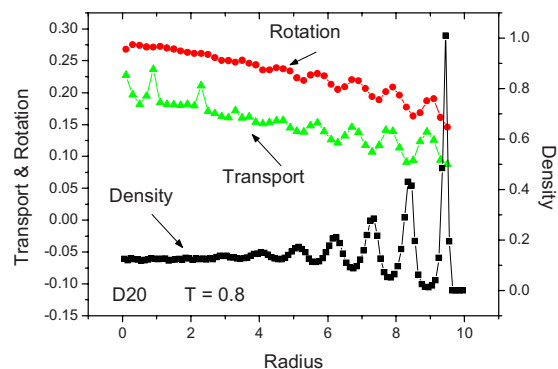


FIG. 7. (Color online) The radial profile of the spatially resolved density ρ (squares), the short-time angular rotation $\theta(\tau^*)$ (circles), and the short-time translation coefficient $D(\tau^*)$ (triangles) of the GB system confined in D20 at $T=0.8$. The time step is set to $\tau^*=0.6$. The center of the pore is located at $r=0$ and the pore wall surface at about $r=10$.

$$\theta(\tau^*) = \frac{2}{\pi} \arccos(C(\tau^*)) = \frac{2}{\pi} \arccos(\langle \mathbf{u}(\tau^* + t_0) \mathbf{u}(t_0) \rangle), \quad (9)$$

and

$$D(\tau^*) = \frac{1}{6\tau^*} \langle |\mathbf{r}(\tau^* + t_0) - \mathbf{r}(t_0)|^2 \rangle. \quad (10)$$

In these equations, $\mathbf{u}(t_0)$ is the vector of the particle principle axis at step t_0 and the position of the particle center at time t_0 is defined as $\mathbf{r}(t_0)$. These two quantities have been derived from the normalized average angular displacement of the molecule axis and the mean-square displacement of the molecular center after a time period τ^* . The time step has been set to a rather small value $\tau^*=0.6$ in order to avoid the particles diffusing from one layer to another during the period from t_0 to $t_0 + \tau^*$. Therefore, the two dynamics parameters can be considered as spatially resolved indicators of the short-time MD. The radial distance intervals were set to 0.1 for the density and 0.2 for the two dynamics parameters, respectively.

Figure 7 shows large oscillations in the density profile near the pore wall. This pseudoperiodic modulation of the local density propagates into the pore volume but is damped and progressively disappears. The typical distance between the two maxima is about σ_0 , which is the side-by-side contact distance of the GB molecules. This feature is a clear evidence of a layering effect and already observed for LCs in slit pores.^{34,37,55}

The radial profile of the short-time angular rotation $\theta(\tau^*)$ clearly shows that the dynamics is not homogeneous over the entire pore volume also. Close to the pore wall, $\theta(\tau^*)$ presents spatial oscillations, which are in antiphase with respect to that of the density profile [$\theta(\tau^*)$ is smaller within the layers of maximum density]. The amplitude of these pseudoperiodic oscillations is small and they are rapidly damped away from the pore surface. This phenomenon is a direct consequence of density modulations. In addition, a gradual increase in $\theta(\tau^*)$ by a factor of 1.5 is observed when moving from the pore wall to the pore center. It is noticeable that $\theta(\tau^*)$ does not saturate at the pore center. It implies that the length of the surface effect virtually exceeds the pore size. A

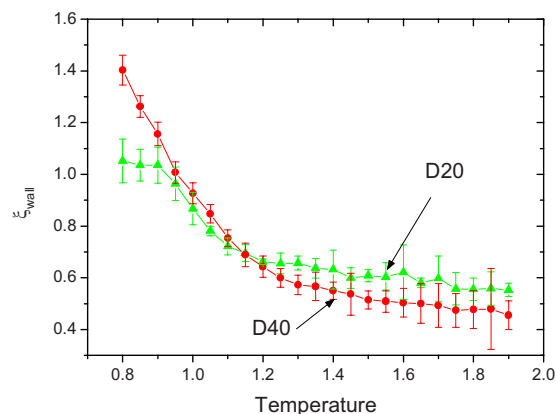


FIG. 8. (Color online) Variation with the temperature of the decay length ξ_{wall} of the density for the GB systems D20 (circles) and D40 (triangles).

very similar behavior is observed in Fig. 7 for the short-time translation coefficient $D(\tau^*)$. It indicates a comparable mechanism for the propagation of surface effects on short time rotational and translational dynamics. Note that this assumption has been introduced in a recent model of QENS experiments on a nanoconfined LC.¹⁶

According to the analyses above, we show that the density, rotation, and translation are deeply influenced by the pore surface. We do not observe two distinct contributions, which might arise from the in-pore system and a separated surface layer, as reported in some experimental cases.^{6,8,9} At variance, with the set of parameters used to describe the LC-wall interaction, the influence of the surface effect gradually declines for molecules located farther from the pore wall, which corresponds to a situation reported for other systems.^{16,52} In order to quantify the thickness of this influenced region, we defined another parameter—the decay length of the density ξ_{wall} . Indeed, the local density presents large variation across the pore diameter. It therefore provides a convenient estimation of ξ_{wall} , although it is not excluded that static and dynamical properties might be propagated differently from the surface. The value of ξ_{wall} was obtained by assuming that the density maxima of the pseudoperiodic oscillation decay exponentially as a function of the distance from the wall according to

$$\rho(r_i) = \rho_0 + [\rho_{\text{wall}} - \rho_0] \exp\left(-\frac{r_i - R_{\text{pore}}}{\xi_{\text{wall}}}\right), \quad (11)$$

where $\rho(r_i)$ corresponds to the density maximum of each layer i located at the distance r_i from the pore center and R_{pore} the pore radius. In Eq. (11), the values of the density at the pore center and the pore wall (extrapolated value) are $\rho_0=0.153$ and ρ_{wall} , respectively. Several layers (up to $i=7$) have been considered to extract ξ_{wall} from an exponential regression of the density according to Eq. (11).

The temperature variation of the density decay length ξ_{wall} for D20 and D40 are shown in Fig. 8. The values vary from 0.5 to 1.4 over the entire temperature range. First it should be noted that these apparently small values do not mean that surface influences affect only one or two molecular layers. In fact, it reflects the typical distance over which the amplitude of density modulation decays by a factor e^1

from its maximum value at the pore wall. For example, ξ_{wall} is 1.1 for the LCs in *D20* at $T=0.8$. However its layering is still visible at distances up to $7\sigma_0$ from the surface as shown in Fig. 7.

In Fig. 8, ξ_{wall} displays a rather similar temperature variation for the two cylinders. At high temperature, ξ_{wall} is insensitive to the temperature and slightly increases from $T=1.9$ to $T=1.3$. However at lower temperature ($T<1.3$), ξ_{wall} shows a notable curvature and rises progressively. This temperature region is accompanied by the appearance of smectic correlations as seen in Fig. 4. The length for *D40* increases to a value of $\xi_{\text{wall}}=1.4$ at the lowest obtained temperature. More interestingly, the decay length saturates to about $\xi_{\text{wall}}=1.1$ in the case of *D20* for temperature lower than $T=0.9$. This effect might be exacerbated by the curvature of the cylindrical pore, which increases the average molecule-surface interaction with respect to a flat surface. It reveals possible cutoff or finite-size effects that are still not present in the largest pore size, which primary reflects surface effects in the studied temperature interval.

VII. CONCLUSION

We compared the properties of the LCs in the bulk and the two cylindrical nanopores confinements (*D20*, *D40*) by using MD simulation method. The simulations were performed as a function of temperature in the range from $T=0.8$ to $T=1.9$ along the isochoric line at $\rho=0.153$. The fluid-wall interactions were introduced as the interaction between the GB (4.4, 20, 1, 1) particles and the close-packed array of LJ (12-6) particles.

The structural properties of the three GB systems were characterized by the orientational and translational order parameters. In confinement, our data revealed that the orientational ordering occurs as a continuous evolution from a high temperature paranematic state to a nematic phase over the entire studied temperature range. This result is in qualitative agreement with previous experiments in porous cylinder and simulation studies on slit pores.^{14,39} The translational ordering is heavily inhibited in confinement. The LC confined in *D20* adopts an homogeneous planar-axial configuration, whereas *D40* forms a planar-tangential structure. This different state is made possible in the present study, which considered the case of a very large pore size *D40*. Within the range of external parameters considered in this work, *D40* apparently illustrates the case of very large pore when surface curvature has small impact on the properties of the confined system. The possible remaining degree of metastability of the planar-tangential structure with respect to a homogeneous planar-axial configuration has not been addressed in this study.

The MD was characterized by the rotational correlation time of the principal axis of the GB particles and the mass center diffusion coefficient. In the bulk, the rotational correlation time presents two distinct regimes above and below the *I-N* transition temperature with two nearly Arrhenius temperature dependences. At variance, the rotational dynamics of the confined GB systems exhibit almost constant activation energy over the entire studied temperature range. At

the same time, the reorientation dynamics of the confined GB particles is much slower than the bulk at high temperature. These phenomena originate from the surface interaction and the paranematic ordering. At lower temperature, the correlation time in the bulk is finally longer than that of the confined phase, because of the intervening *N-SmA* phase transition, which is suppressed in confinement.

The translational diffusion appears to be sensitive to the translational order rather than the orientational order. In the bulk GB system, a crossover between two almost Arrhenius behaviors is observed when approaching the *N-SmA* transition. The diffusion coefficients of the confined GB systems show a smooth bend in the same temperature range as a reminiscent of the bulk transition and related to the occurrence of end-to-end molecular correlations. The molecular translation of the LCs in confinement is significantly slowed down by the pore surface at high temperature, whereas this effect is balanced by the absence of long-range order at lower temperature. As a result, the translational dynamics of the confined phases is eventually faster than the bulk at low temperature and similar to that of the rotation.

Finally, we investigated the in-pore properties of the confined GB systems. The properties are generally inhomogeneous. Strong layering effects are induced by the surface and propagate into the pore volume. The thickness of this interfacial region increases more rapidly at low temperature. Up to seven layers are formed below $T=1.0$ and possible finite-size effects are observed for the small cylinder. Spatially resolved translational diffusion coefficient and main axis rotation amplitude show that the dynamics within the contact layer is strongly hindered. At variance to some experimental observations, the contact layer does not lead to a distinct relaxation process, well separated from a bulklike dynamics in the center of the pore.^{6,8,9} On the contrary, the hindrance is gradually transmitted from the pore wall to the confined phase. It leads to a spatially heterogeneous dynamics at the nanometer scale, which is in qualitative agreement with recent experiments.^{15,16} The simulation study provides a direct insight into the microscopic origin of the glasslike and non-Debye characters of the self-particle dynamics of LCs in nanochannels.^{6,8,9,16}

This study corresponds to the first stage of a series of simulation studies. As inspired by recent experimental results, the following investigation will ultimately address pore topology and surface roughness effects. Another extension of the study is also in progress in order to avoid the limitation of constant density simulations and to check the possible occurrence of a short-range ordered smectic phase at lower temperature as reported by experiments in cylindrical pores.¹³

ACKNOWLEDGMENTS

This study was part of the post-doctoral fellowship project of Q.J., supported by a grant from the CNRS. R.B. benefited from a doctoral research grant from the Brittany Region. Financial support from the Centre de Compétence C'Nano Nord-Ouest is expressly acknowledged.

- ¹L. Crystals, in *Complex Geometries*, edited by G. P. Crawford and S. Zumer (Taylor and Francis, London, 1996).
- ²G. P. Crawford, R. Stannarius, and J. W. Doane, *Phys. Rev. A* **44**, 2558 (1991).
- ³G. P. Crawford, D. W. Allender, and J. W. Doane, *Phys. Rev. A* **45**, 8693 (1992).
- ⁴G. S. Iannacchione, G. P. Crawford, S. Žumer, J. W. Doane, and D. Finotello, *Phys. Rev. Lett.* **71**, 2595 (1993); G. S. Iannacchione and D. Finotello, *Phys. Rev. E* **50**, 4780 (1994).
- ⁵T. Bellini, L. Radzihovsky, J. Toner, and N. A. Clark, *Science* **294**, 1074 (2001).
- ⁶G. P. Sinha and F. M. Aliev, *Phys. Rev. E* **58**, 2001 (1998).
- ⁷F. M. Aliev, *J. Non-Cryst. Solids* **307–310**, 489 (2002).
- ⁸L. Frunza, S. Frunza, H. Kosslick, and A. Schönals, *Phys. Rev. E* **78**, 051701 (2008).
- ⁹S. Frunza, L. Frunza, H. Goering, H. Sturm, and A. Schönals, *Europhys. Lett.* **56**, 801 (2001).
- ¹⁰R. L. Leheny, S. Park, R. J. Birgeneau, J.-L. Gallani, C. W. Garland, and G. S. Iannacchione, *Phys. Rev. E* **67**, 011708 (2003).
- ¹¹P. Kumar, T. Hofmann, K. Knorr, P. Huber, P. Scheib, and P. Lemmens, *J. Appl. Phys.* **103**, 024303 (2008).
- ¹²R. Guégan, D. Morineau, C. Loverdo, W. Béziel, and M. Guendouz, *Phys. Rev. E* **73**, 011707 (2006).
- ¹³R. Guégan, D. Morineau, R. Lefort, W. Béziel, M. Guendouz, L. Noirez, A. Henschel, and P. Huber, *Eur. Phys. J. E* **26**, 261 (2008).
- ¹⁴A. V. Kityk, M. Wolff, K. Knorr, D. Morineau, R. Lefort, and P. Huber, *Phys. Rev. Lett.* **101**, 187801 (2008).
- ¹⁵R. Guégan, D. Morineau, R. Lefort, A. Moréac, W. Béziel, M. Guendouz, J.-M. Zanotti, and B. Frick, *J. Chem. Phys.* **126**, 064902 (2007).
- ¹⁶R. Lefort, D. Morineau, R. Guégan, M. Guendouz, J.-M. Zanotti, and B. Frick, *Phys. Rev. E* **78**, 040701 (2008).
- ¹⁷C. M. Care and D. J. Cleaver, *Rep. Prog. Phys.* **68**, 2665 (2005).
- ¹⁸J. G. Gay and B. J. Berne, *J. Chem. Phys.* **74**, 3316 (1981).
- ¹⁹G. R. Luckhurst, R. A. Stephens, and R. W. Phippen, *Liq. Cryst.* **8**, 451 (1990).
- ²⁰E. de Miguel, L. F. Rull, M. K. Chalam, K. E. Gubbins, and F. van Swol, *Mol. Phys.* **72**, 593 (1991).
- ²¹E. de Miguel, *Mol. Phys.* **100**, 2449 (2002).
- ²²E. de Miguel, L. F. Rull, and K. E. Gubbins, *Phys. Rev. A* **45**, 3813 (1992).
- ²³P. P. Jose, D. Chakrabarti, and B. Bagchi, *Phys. Rev. E* **71**, 030701 (2005); D. Chakrabarti and B. Bagchi, *J. Phys. Chem. B* **111**, 11646 (2007).
- ²⁴G. R. Luckhurst and P. S. J. Simmonds, *Mol. Phys.* **80**, 233 (1993).
- ²⁵M. A. Bates and G. R. Luckhurst, *J. Chem. Phys.* **110**, 7087 (1999).
- ²⁶E. de Miguel, E. M. del Río, and F. J. Blas, *J. Chem. Phys.* **121**, 11183 (2004).
- ²⁷E. de Miguel, F. J. Blas, and E. M. del Río, *Mol. Phys.* **104**, 2919 (2006).
- ²⁸K. Satoh, *J. Chem. Phys.* **124**, 144901 (2006).
- ²⁹M. A. Bates and G. R. Luckhurst, *Mol. Phys.* **99**, 1365 (2001).
- ³⁰M. A. Bates and G. R. Luckhurst, *J. Chem. Phys.* **120**, 394 (2004).
- ³¹K. Satoh, *J. Chem. Phys.* **125**, 204902 (2006).
- ³²Z. Zhang, A. Chakrabarti, O. G. Mouritsen, and M. J. Zuckermann, *Phys. Rev. E* **53**, 2461 (1996).
- ³³J. Stelzer, P. Galatola, G. Barbero, and L. Longa, *Phys. Rev. E* **55**, 477 (1997).
- ³⁴T. Gruhn and M. Schoen, *Phys. Rev. E* **55**, 2861 (1997); *J. Chem. Phys.* **108**, 9124 (1998).
- ³⁵V. Palermo, F. Biscarini, and C. Zannoni, *Phys. Rev. E* **57**, R2519 (1998).
- ³⁶G. D. Wall and D. J. Cleaver, *Mol. Phys.* **101**, 1105 (2003).
- ³⁷M. K. Chalam, K. E. Gubbins, E. de Miguel, and L. F. Rull, *Mol. Simul.* **7**, 357 (1991).
- ³⁸G. D. Wall and D. J. Cleaver, *Phys. Rev. E* **56**, 4306 (1997).
- ³⁹E. Gwóźdź, K. Pasterny, and A. Bródka, *Chem. Phys. Lett.* **329**, 106 (2000); **335**, 71 (2001).
- ⁴⁰J. Quintana, E. C. Poiré, H. Domínguez, and J. Alejandro, *Mol. Phys.* **100**, 2597 (2002).
- ⁴¹D. L. Cheung and F. Schmid, *J. Chem. Phys.* **122**, 074902 (2005); *Chem. Phys. Lett.* **418**, 392 (2006).
- ⁴²D. de las Heras, E. Velasco, and L. Mederos, *Phys. Rev. Lett.* **94**, 017801 (2005); *Phys. Rev. E* **74**, 011709 (2006).
- ⁴³T. Bellini, C. Chiccoli, P. Pasini, and C. Zannoni, *Phys. Rev. E* **54**, 2647 (1996).
- ⁴⁴Yu. Trukhina and T. Schilling, *Phys. Rev. E* **77**, 011701 (2008).
- ⁴⁵A. P. J. Emerson and C. Zannoni, *J. Chem. Soc., Faraday Trans.* **91**, 3441 (1995).
- ⁴⁶T. Mima and K. Yasuoka, *Phys. Rev. E* **77**, 011705 (2008).
- ⁴⁷S. J. Plimpton, *J. Comput. Phys.* **117**, 1 (1995).
- ⁴⁸K. Satoh, S. Mita, and S. Kondo, *Chem. Phys. Lett.* **255**, 99 (1996).
- ⁴⁹R. Latham and D. J. Cleaver, *Chem. Phys. Lett.* **330**, 7 (2000).
- ⁵⁰Z. Kutnjak, S. Kralj, G. Lahajnar, and S. Žumer, *Phys. Rev. E* **68**, 021705 (2003).
- ⁵¹R. Zorn, L. Hartmann, B. Frick, D. Richter, and F. Kremer, *J. Non-Cryst. Solids* **307–310**, 547 (2002).
- ⁵²P. Scheidler, W. Kob, and K. Binder, *Eur. Phys. J. E* **12**, 5 (2003).
- ⁵³D. Morineau, Y. Xia, and C. Alba-Simionesco, *J. Chem. Phys.* **117**, 8966 (2002).
- ⁵⁴C. Alba-Simionesco, G. Dosseh, E. Dumont, B. Frick, B. Geil, D. Morineau, V. Teboul, and Y. Xia, *Eur. Phys. J. E* **12**, 19 (2003); Y. Xia, G. Dosseh, D. Morineau, and C. Alba-Simionesco, *J. Phys. Chem. B* **110**, 19735 (2006).
- ⁵⁵R. Guégan, D. Morineau, and C. Alba-Simionesco, *Chem. Phys.* **317**, 236 (2005).
- ⁵⁶D. Morineau, R. Guégan, Y. Xia, and C. Alba-Simionesco, *J. Chem. Phys.* **121**, 1466 (2004).